mibelle GROUP biochemistry

in-cosmetics global
Innovation Zone Best Ingredient Award 2025
Active Ingredient SHORTLISTED

AmelioSense™

Calm the flame of pyroptosis – soothe the irritated skin

SWISS QUALITY PRODUCT

AmelioSense™

Calm the flame of pyroptosis – soothe irritated skin

Combined Power Against Redness and Inflammation
AmelioSense™ is a comprehensive approach aimed at effectively combating skin irritation and redness. AmelioSense™ combines the extract of shepherd's purse with a liposomal preparation that contains three powerful antioxidants. AmelioSense™ works effectively against pyroptosis, a programmed, highly inflammatory cell death, while at the same time reducing general skin inflammation processes that lead to redness and irritated skin.

Shepherd's purse, which originally grew in the Mediterranean region, is an ancient remedy that is used to reduce inflammatory skin conditions. Mibelle Biochemistry has put it together with a strong antioxidant complex, which combines carnosine with tocopherol and silymarin from milk thistle extract to effectively combat redness and irritated skin in the face.

In vitro and *in vivo* studies have demonstrated that AmelioSense™:
- inhibits active caspase-1, a key enzyme in pyroptosis
- reduces the release of free radicals
- reduces the gene expression of diverse inflammatory markers known to be present in rosacea
- visibly reduces the appearance of blood vessels and redness in the cheek area

AmelioSense™ is a promising new active ingredient for skin application that is particularly suitable for treating inflamed, irritated and flushed skin.

Watch the product movie here!

AmelioSense™
- Reduces the appearance of visible blood vessels on the cheeks
- Decreases facial redness and blood flow
- Alleviates irritation
- Promotes a calm and balanced skin tone

Applications
- Soothing anti-redness serum for delicate skin
- Redness relief soothing cream
- Ultra calming anti-redness mask
- Instant relief barrier strengthening serum
- Redness relief daily cleansing milk

Formulating with AmelioSense™
- Recommended use level: 2%
- Incorporation: For cold processes, dissolve AmelioSense™ into the aqueous phase. In hot/cold processes, add during the cooling phase below 40°C.
- Thermostability: Temperatures of up to 60°C for a short time will not affect the stability of AmelioSense™.

INCI (EU/PCPC) Declaration
Capsella Bursa-Pastoris Extract (and) Lecithin (and) Carnosine (and) Tocopherol (and) Silybum Marianum Fruit Extract (and) Maltodextrin (and) Aqua/Water

Rosacea
A complex inflammatory process

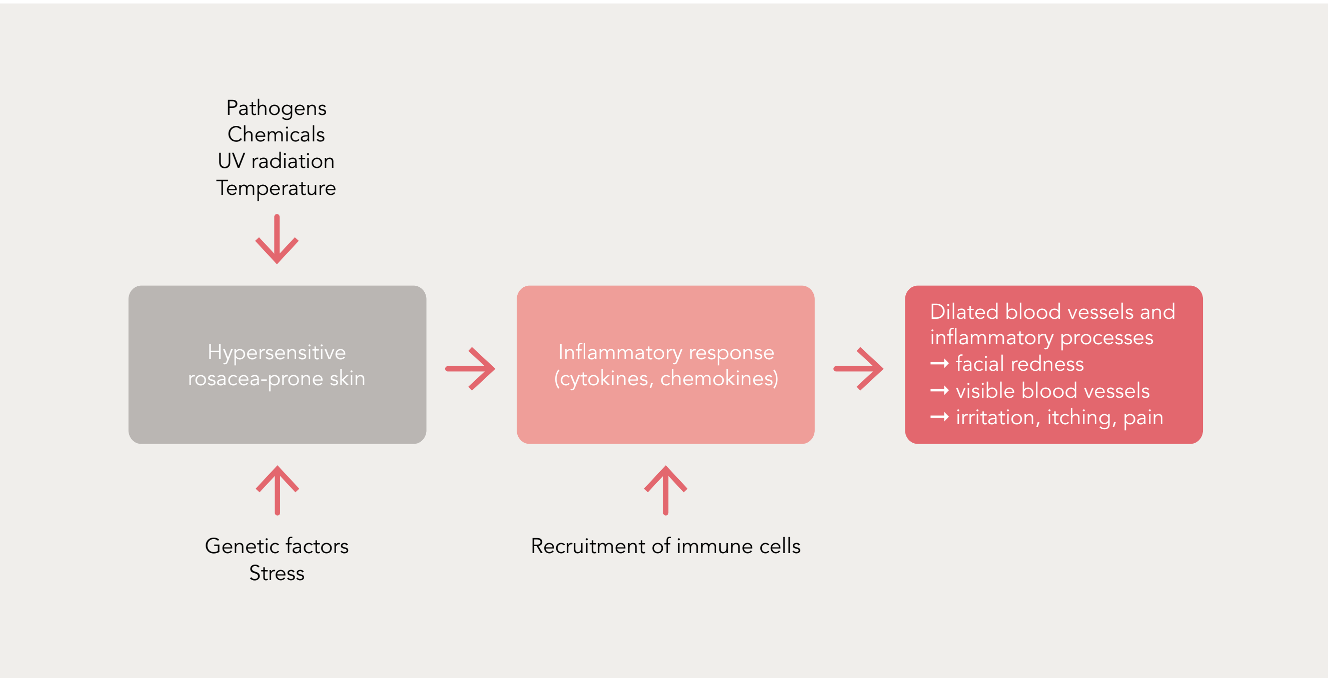
Rosacea is Characterized by Redness and Visible Blood Vessels in the Face
Rosacea is a very complex chronic inflammatory skin condition that primarily affects the face and, in particular, the cheek and chin areas. Inflammation plays a central role in the development and progression of this condition. The most visible symptoms of rosacea include skin redness and visible blood vessels, and patients can also sometimes experience itching and pain. Despite its high prevalence, the causes and promoters of rosacea are still not fully understood. However, it is likely to be multifactorial, involving both genetic and environmental factors.

Patients with this condition often experience a significant reduction in terms of quality of life, self-esteem, and general wellbeing. However, the successful treatment of this condition can be challenging, and it usually involves topical anti-inflammatories, antibiotics, and sunscreens.

Pro-Inflammatory Mediators are Elevated in Rosacea-Prone Skin
Rosacea can be triggered by numerous factors, including chemical stimuli, environmental factors, pathogens, and genetic predispositions. Therefore, the pathophysiology of rosacea patients is extremely complex.

The skin of rosacea patients is hypersensitive to external stimuli (such as UV, temperature, lesions, etc.), which leads to an increased inflammatory response that is mediated by the release of pro-inflammatory mediators such as various cytokines and chemokines. These molecules recruit immune cells (mast cells) and contribute to the amplification of the inflammatory response as well as to increased blood vessel formation. The combination of dilated blood vessels and chronic inflammation causes the typical symptoms of rosacea, such as redness in the face.

Causes and Effects of Rosacea



Pyroptosis
A spreading fire that can worsen the symptoms of rosacea

Pyroptosis is a term that is derived from the Greek words *pyro* (fire) and *ptosis* (falling). It vividly translates to "fiery falling," which captures the explosive release of pro-inflammatory signals from a cell in its death throes. Unlike other forms of programmed cell death, pyroptosis ignites a rapid and intense inflammatory response, which is crucial for the body's defense system against infection.

Upon detecting pathogens, specific receptors on the skin cell trigger the formation of a large protein complex that is known as the inflammasome. This complex activates caspase-1, which is the key enzyme of pyroptosis. Caspase-1 then processes and unleashes potent pro-inflammatory mediators such as interleukin IL-18 and causes the formation of reactive oxygen species (ROS). Caspase-1 also orchestrates the formation of pores in the cell membrane, which leads to the catastrophic loss of cell integrity and eventual cell rupture. The ensuing release of cellular contents and inflammatory mediators summons immune cells to the site, which sparks a robust immune response. However, these released components can also harm nearby uninfected cells.

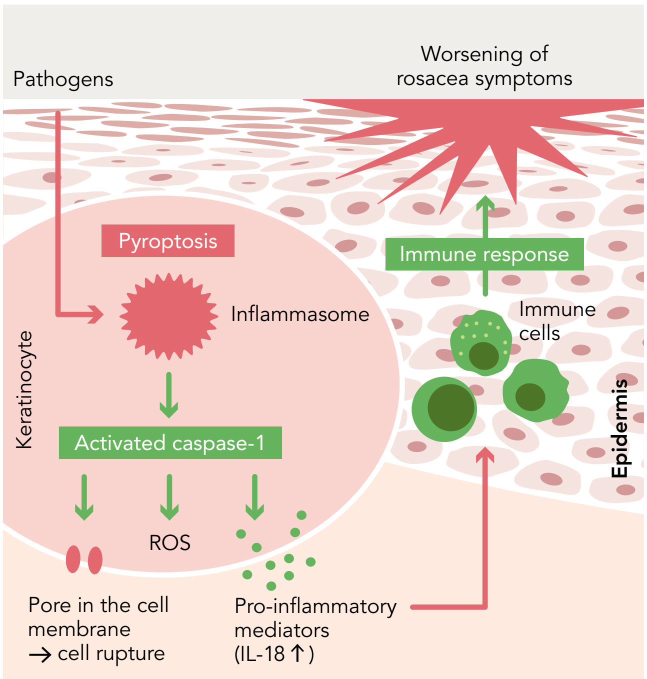
Therefore, while pyroptosis serves as a rapid defense mechanism, it can also "spread the fire," which can lead to chronic inflammation and tissue damage if over-activated.

The Interplay Between Pyroptosis and Rosacea
Recent research has unveiled a critical link, which shows that pyroptosis can exacerbate rosacea (1). The activation of the inflammasome and the subsequent pyroptotic processes are central to the chronic inflammation characteristic of rosacea. The pro-inflammatory mediators that are released during pyroptosis promote the infiltration, activation, and differentiation of immune cells, which heightens the immune and inflammatory response and worsens rosacea symptoms. Therefore, targeting pyroptosis is essential for optimizing care of rosacea and inflammation.

With AmelioSense™, Mibelle Biochemistry has found a new approach to effectively fight pyroptosis and inflammation.

(1) Hu et al. (2023) Experimental dermatology 33(1): e14812.

Pyroptosis Can Worsen Rosacea



Shepherd's Purse & Antioxidant Complex

The story of a strong anti-inflammatory combination

Shepherd's Purse – A Versatile Medicinal Herb Featuring Anti-Pyroptotic Properties

Shepherd's purse comes from the cruciferous family *(Brassicaceae)*. The scientific name *Capsella bursa-pastoris* is made up of the Latin words *capsa* for "capsule," *bursa* for "pocket," and *pastor* for "shepherd." It refers to the fact that the little flat green fruits of the plant are shaped like the pockets of former shepherds. This remarkable plant has a long history in natural medicine. In ancient times, shepherd's purse was valued for its potential to treat skin conditions. The plant has anti-inflammatory effects and promotes wound healing, which makes it a popular household remedy.

Based on these known indications and effects, the extract from shepherd's purse was tested for its efficacy in terms of combating pyroptosis and it proved its effect in this regard.

The Perfect Addition: A Powerful Antioxidant Complex

Another component of AmelioSense™ is a complex that combines the potent antioxidant carnosine with the two powerful, oil-soluble antioxidants tocopherol and silymarin, which is extracted from the milk thistle. It is referred to here as an "antioxidant complex." All three antioxidants offer unique benefits.

Carnosine, which is a dipeptide, primarily fights skin glycation and oxidative stress. Silymarin is a flavonoid that is known for its anti-inflammatory properties, while tocopherol is a powerful antioxidant that protects against oxidative damage. This antioxidant complex is a liposomal formulation that uses lecithin to improve the penetration of these ingredients into the skin.

The combination of shepherd's purse extract and antioxidant complex was spray-granulated onto maltodextrin to produce AmelioSense™. Mibelle Biochemistry has harnessed the calming properties of shepherd's purse and the protective effects of the antioxidant complex, combining them into AmelioSense™, an innovative formulation that offers remarkable benefits for the fight against inflammation.

In a specific skin model, this active ingredient significantly reduced inflammation in skin cells and effectively combats skin redness and the appearance of blood vessels.

Shepherd's Purse *(Capsella Bursa-Pastoris)*

AmelioSense™
Study results

Inhibitory Effect on Active Caspase-1
Shepherd's purse extract was tested for its effect on pyroptosis in normal human epidermal keratinocytes (NHEK). A pyroptosis trigger, high-mobility group protein B1 (HMGB1), was used to activate the inflammasome, which then activates the key enzyme caspase-1 as an important step in the pyroptosis pathway.

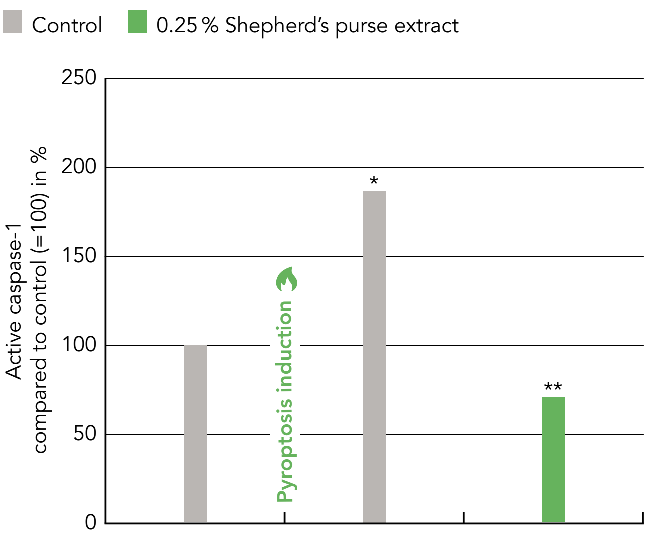
Cells were pretreated for a period of 24 hours with 0.25 % shepherd's purse extract or not (control). The medium was then replaced by medium containing HMGB1 (either alone or together with the extract) for a further 24 hours to trigger pyroptosis in these cells.

Active caspase-1 was then detected by fluorescence microscopy. Furthermore, cell nuclei were stained with Hoechst to normalize active caspase-1 to cell numbers.

Results showed that active caspase-1 levels increased after the addition of HMGB1, which confirmed that pyroptosis was induced. Treatment with 0.25 % shepherd's purse extract significantly reduced active caspase-1 levels even below those of untreated, non-stressed cells.

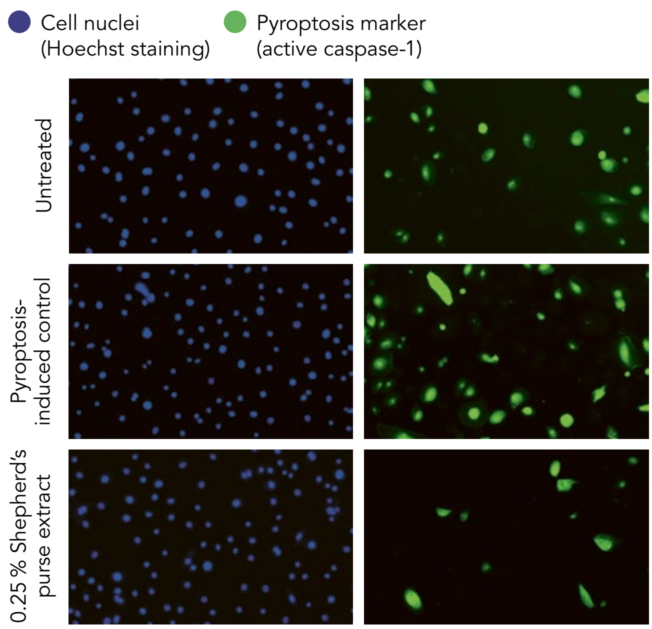
To summarize, these results indicate that shepherd's purse has a significant inhibitory effect on pyroptosis by reducing active caspase-1 as an important enzyme in the pyroptosis pathway. Therefore, AmelioSense™ could stop the spreading of the inflammation and thus has the potential to soothe irritated, inflamed skin.

Inhibition of the Pyroptosis Key Enzyme Caspase-1



*p<0.05 versus untreated control
**p<0.05 versus pyroptosis-induced control

S-889 / © Mibelle Biochemistry



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AmelioSense™
Study results

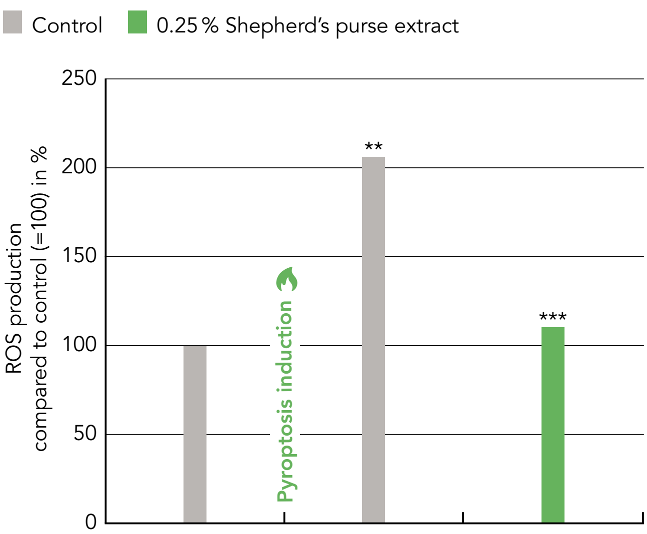
Reduction in ROS Release
After triggering inflammasome-induced pyroptosis in normal human epidermal keratinocytes (NHEK) with HMGB1, the release of reactive oxygen species (ROS) was evaluated. Reactive oxygen species are free radicals, and they can potentially damage the cells.

NHEK were either pretreated for a period of 24 hours with 0.25 % shepherd's purse extract or not pretreated (the control). The medium was then replaced by medium containing HMGB1 (either alone or together with the extract) as well as CellROX green reagent, to visualize ROS, and the cells were then incubated for an additional hour.

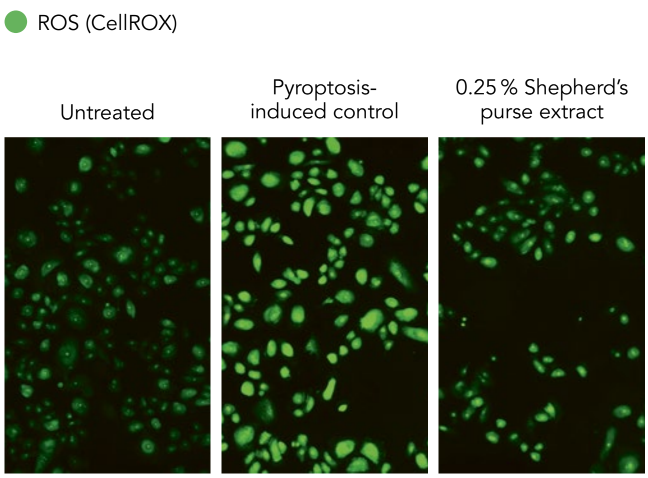
Cells were then fixed, and the fluorescence intensity as a read-out for oxidative stress and formation of ROS was measured.

Results showed that HMGB1 strongly increases ROS production in keratinocytes, and treatment with 0.25 % shepherd's purse extract significantly reduced HMGB1-induced ROS release almost back to unstressed levels. The potential to reduce ROS release in a pyroptosis-triggered environment showed that AmelioSense™ reduces oxidative stress and thus could lead to a calmer skin.

Reduction of ROS



**p<0.01 versus untreated control
***p<0.01 versus pyroptosis-induced control



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Reduction in IL-18 Gene Expression

The pro-inflammatory cytokine interleukin (IL)-18 is converted into its active form by the enzyme caspase-1, and it contributes decisively to the inflammatory response and the clinical symptoms of rosacea, such as redness and skin lesions.

IL-18 expression was evaluated after HMGB1-induced pyroptosis. For this, NHEK were either pretreated with 0.25 % shepherd's purse extract or not pretreated (the control) for a period of 24 hours. The medium was replaced by medium containing HMGB1 (either alone or together with the extract) and incubated for 6 hours. Following this, gene expression analysis was performed via RT-qPCR.

Results showed that HMGB1 stress increases IL-18 levels. Pretreatment with shepherd's purse extract was able to reduce HMGB1-induced IL-18 gene expression levels back to unstressed levels.

Reduced IL-18 means that less inflammation is triggered in the skin.

In conclusion, by reducing IL-18 gene expression, ROS-release, and active caspase-1, shepherd's purse extract can reduce the stress-induced pyroptosis signaling pathway.

Therefore, AmelioSense™ could strongly contribute to combating inflammation and redness in the skin.

Reduction in IL-18-Expression

AmelioSense™
Study results

Reduction of Inflammatory Markers

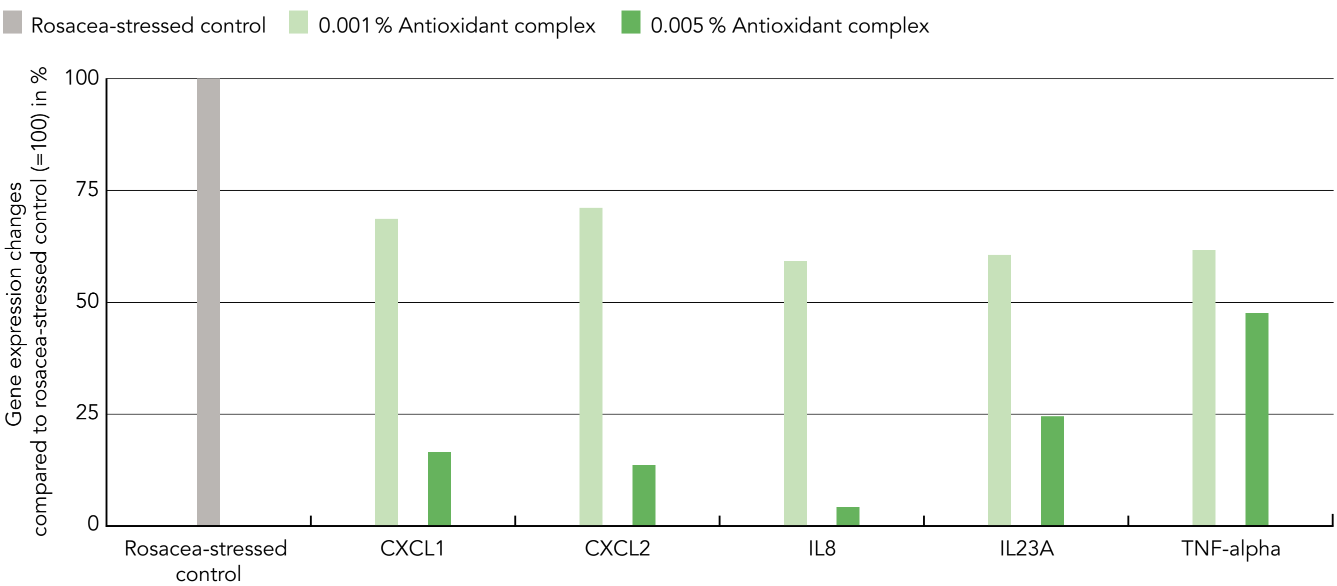
In this assay, the antioxidant complex, which is a component of AmelioSense™, was tested for its effect on inflammatory markers that are also known to be present in rosacea. The antioxidant complex was tested at 0.001 % and 0.005 %. Normal human epidermal keratinocytes (NHEK) were stimulated with a combination of several trigger factors (e. g. LL-37, Vitamin D, IL-17) to induce an inflamed environment comparable to rosacea.

Cells were either pretreated with the antioxidant complex or not pretreated (the control) for a period of 24 hours. After preincubation, the medium was replaced with an assay medium either containing or not (the control) the antioxidant complex as well as the trigger factors. After 48 hours of incubation, the cells were then harvested, and gene expression was analyzed by RT-qPCR.

The rosacea-like environment enhanced the expression of cytokines and chemokines as well as markers for the innate immunity in keratinocytes. Cotreatment of keratinocytes with the antioxidant complex resulted in a dose-dependent downregulation of inflammatory markers, which are also known to be crucial in rosacea. The gene expression of the chemokines CXCL1, CXCL2, and the cytokines IL8, IL23A, and TNF-alpha was strongly reduced by 83 %, 86 %, 95 %, 75 %, and 52 %, respectively, at a concentration of 0.005 % of the antioxidant complex.

To summarize, these results indicate that the antioxidant complex has a potent anti-inflammatory capacity, which makes AmelioSense™ a powerful active in regard to fighting skin inflammation and irritation.

Reduction of Inflammatory Markers Involved in Rosacea



Combining Shepherd's Purse Extract and Antioxidant Complex Reduces Inflammation

In this assay, the anti-inflammatory effect of the combined shepherd's purse extract and the antioxidant complex (AmelioSense™ suitable for *in vitro* testing) was tested on normal human epidermal keratinocytes (NHEK) challenged with TNF-alpha to induce the release of CXCL5 and MCP-1. These chemokines play an important role in the recruitment of immune cells to sites affected by infection and inflammation, and increased expression can contribute to the worsening of redness, flushing, and a burning sensation.

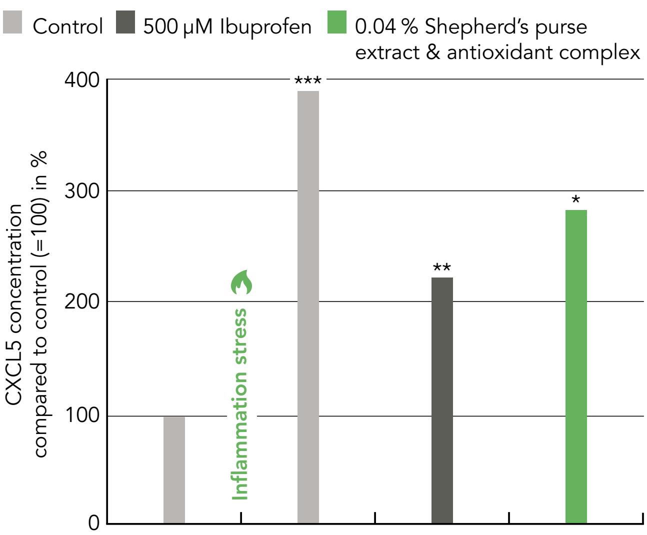
NHEK were pretreated for a period of 2 hours either with the combination of shepherd's purse and antioxidant complex or with ibuprofen as the positive control, or left untreated as the negative control. This pre-treatment was followed by a 24-hour challenge with TNF-alpha in the presence of 0.04 % of the combination or the controls. Meanwhile, one control remained unstressed.

Quantification of CXCL5 and MCP-1 in the culture supernatants at the end of the experiment was carried out using specific ELISA assays.

The results showed that with the combination, the release of CXCL5 after triggering the cells with TNF-alpha was significantly reduced by 105 % compared to the stressed control. MCP-1 was reduced by 266 % compared to the stressed control.

Therefore, this shows that the combined shepherd's purse extract and antioxidant complex can effectively combat inflammation and thus contribute to skin soothing.

Reduction of Inflammation (CXCL5)



*p<0.05 versus control and versus challenged control
**p<0.01 versus control and versus challenged control
***p<0.001 versus control

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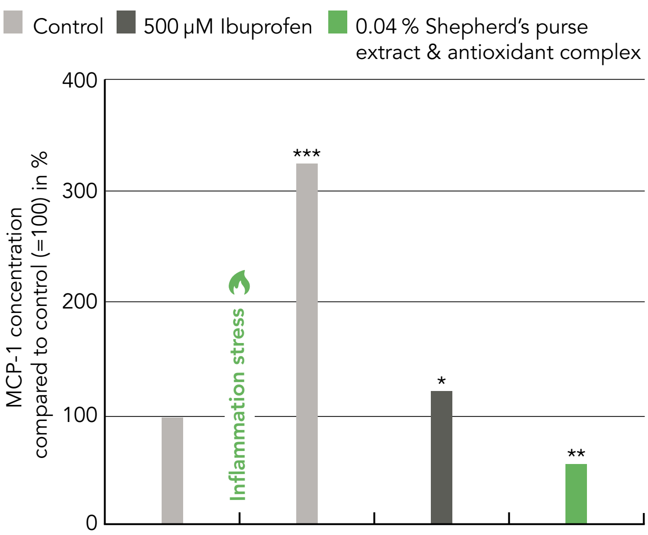
Reduction of Inflammation (MCP-1)



*p<0.05 versus control and versus challenged control
**p<0.01 versus control and versus challenged control
***p<0.001 versus control

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AmelioSense™
Study results

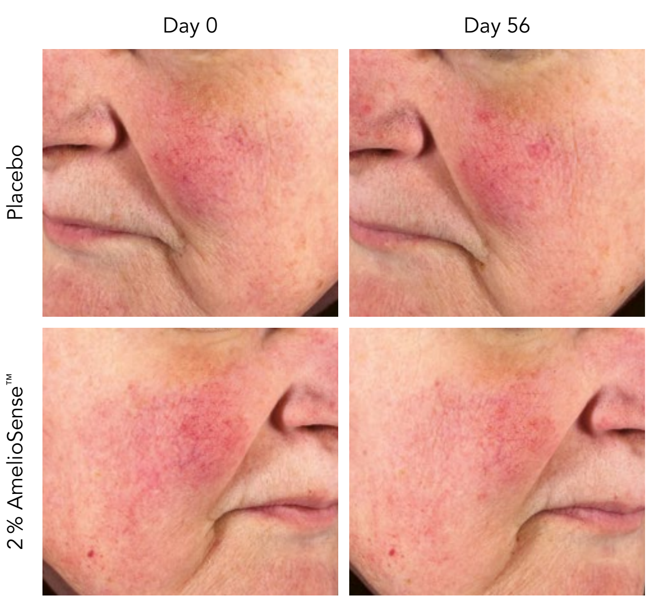
Visible Improvement of Skin Redness
In this double-blind, randomized, placebo-controlled clinical trial, the efficacy of AmelioSense™ in terms of improving skin redness was investigated. Twenty-two male and female volunteers aged between 29 and 70 years (mean age: 54.9 years) who were affected by redness, irritated skin and visible blood vessels in the cheek area were selected. The volunteers applied a cream with 2% AmelioSense™ to one side of their face twice a day and the corresponding placebo cream to the other side.

To investigate the visible results on the skin, high-resolution standardized images were recorded at each time point on the entire face by the ColorFace® system.

As shown in the picture of both halves of the face of the same volunteer below, treatment with 2% AmelioSense™ led to a reduction of visible blood vessels and redness on the face after 56 days. On the other hand, the placebo treatment had no effect at all.

To conclude, AmelioSense™ had a significant impact in terms of reducing visible skin redness, flushes and irritations in the face, which can lead to improved quality of life.

AmelioSense™ Visibly Reduces Skin Redness



Reduction in Redness and Blood Flow

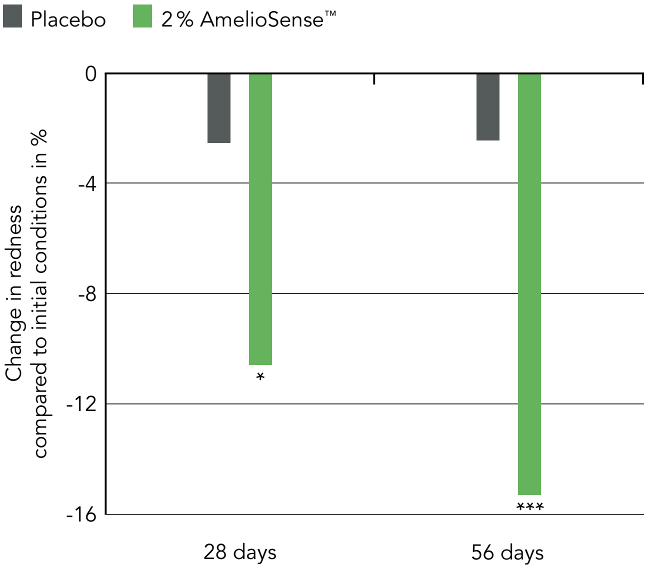
Inflamed skin conditions can be characterized by redness on the face, which is caused by increased blood flow and the permanent, visible dilation of small blood vessels on the skin's surface. Targeting the processes that lead to increased blood flow and the dilation of small blood vessels on the skin's surface can help to alleviate skin flushes and improve the appearance of the skin.

Skin redness was measured using a Spectrophotometer® CM-700d after 28 days and 56 days, and blood flow was measured in tiny blood vessels under the epidermis (capillaries) with a Laser Doppler VMS-LDF2® instrument at the same time points.

After 28 days of treatment with 2% AmelioSense™, redness was significantly reduced by 10.6% and after 56 days it was reduced by 15.3% compared to initial conditions. Furthermore, blood flow was significantly reduced by 29.1% after 28 days and by 46.3% after 56 days compared to initial conditions. Furthermore, all results were statistically significant compared to the placebo treatment.

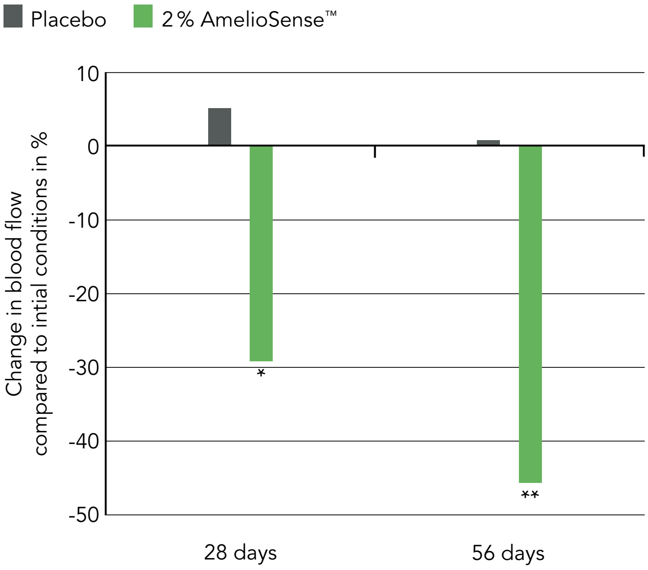
The lower the blood flow, the less severe the skin flushes – AmelioSense™ showed its ability to significantly reduce blood flow and facial redness.

Reduction of Skin Redness



*p<0.05 versus initial conditions and placebo
***p<0.001 versus initial conditions and placebo

Reduction of Blood Flow



*p<0.05 versus initial conditions and placebo
**p<0.01 versus initial conditions and placebo

AmelioSense™
Study results

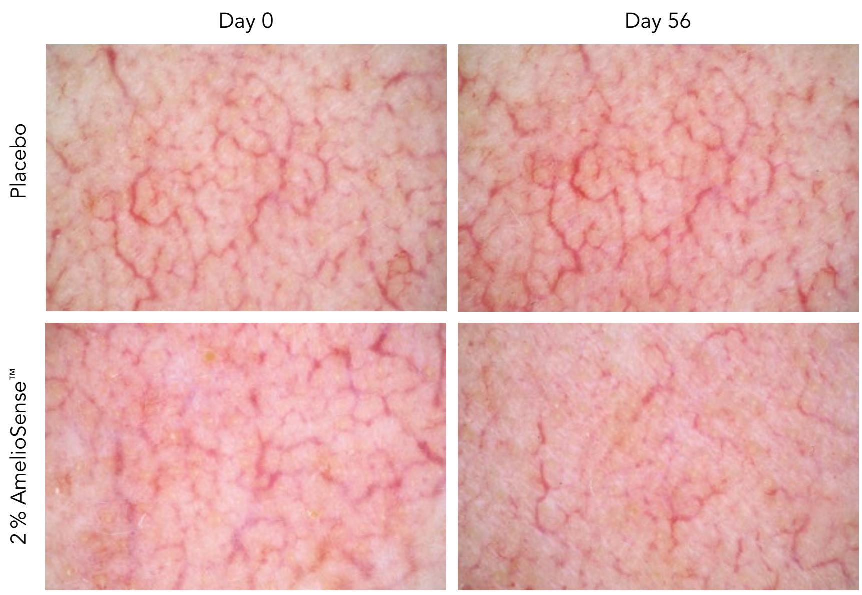
A Zoom-in to the Cheek Region: Reduction of Visible Blood Vessels
In addition to the whole face pictures, the effect of AmelioSense™ on the detailed redness and blood vessel appearance in the cheek region was investigated. For this process, 22 volunteers applied a cream with 2% AmelioSense™ to one side of their faces twice a day and the corresponding placebo cream to the other side for a period of 56 days.

Close-up images of the skin were obtained using a Dermatoscope® Delta20 Plus system mounted on a Canon EOS 450D reflex camera. Images were collected in selected regions of interest displaying visible blood vessels located on each half of the face and image collection was performed after 56 days in the same facial area.

The zoomed images clearly show the visible improvements in terms of blood vessel appearance after 56 days compared to initial conditions and compared to the placebo.

These results suggest that AmelioSense™ helps to improve the appearance of the skin, providing a comprehensive solution for redness and irritation of the skin.

Reduction of Blood Vessel Appearance in the Cheek Area



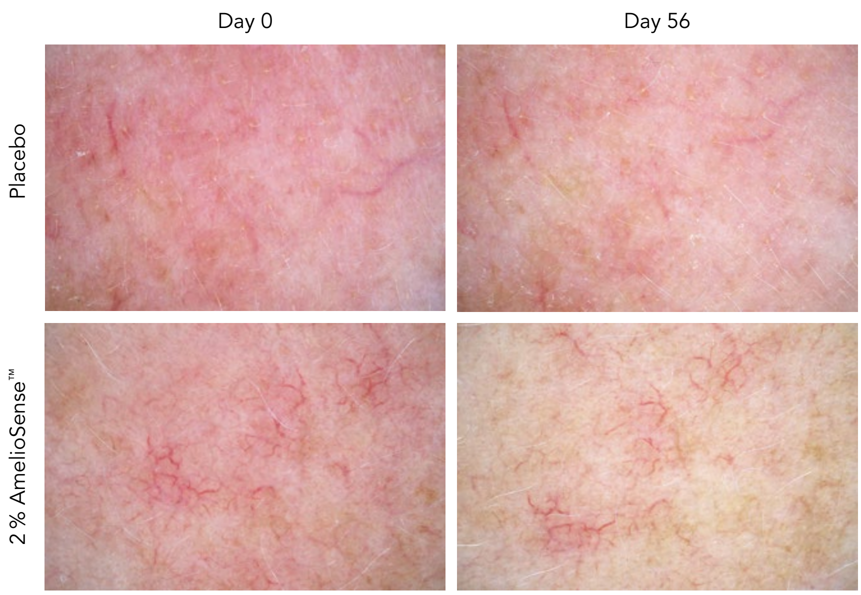
A Zoom-in to the Cheek Region: Reduction of Skin Redness

The close-up pictures investigating redness showed that AmelioSense™ visibly improved skin conditions after 56 days of treatment compared to initial conditions and compared to the placebo.

Moreover, the volunteers self-evaluated AmelioSense™ with the help of a questionnaire at the end of the study. 82 % of the volunteers reported that their skin showed reduced flushing effects. In addition, 91 % of volunteers described the product as having had a soothing effect on their skin.

In summary, AmelioSense™ effectively improves the visible signs of an irritated skin, which consists of skin redness and dilated blood vessels, and provides a soothing feeling for the skin. Therefore, AmelioSense™ helps to improve the quality of life of people suffering from facial redness.

Reduction of Redness in the Cheek Area



AmelioSense™

Calm the flame of pyroptosis – soothe irritated skin

AmelioSense™

- Reduces the appearance of visible blood vessels on the cheeks
- Decreases facial redness and blood flow
- Alleviates irritation
- Promotes a calm and balanced skin tone

Applications

- Soothing anti-redness serum for delicate skin
- Redness relief soothing cream
- Ultra calming anti-redness mask
- Instant relief barrier strengthening serum
- Redness relief daily cleansing milk

in-cosmetics global Innovation Zone Best Ingredient Award 2025 Active Ingredient SHORTLISTED

Marketing Benefits

- Clearly visible effects
- Gentle soothing ingredient to treat irritated and flushed skin
- Targets pyroptosis, a new mechanism that underlies inflammation and rosacea
- Shortlisted for Innovation Zone Best Ingredient at in-cosmetics Global 2025

Innovating for your success

Mibelle Biochemistry designs and develops innovative, high-quality actives based on naturally derived compounds and profound scientific know-how. Inspired by nature – Realized by science.

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